Henry Chang

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A robot **control & planning engineer** bridging **model-based** and **data-driven** method, with 5 yrs Robotics, 3 yrs Autonomous Vehicles, and 3 yrs Intelligent Automation experience, delivering algorithms from concept to production.

University of Washington, PhD in **Mechanical Eng. (Data Science track),** Current GPA 3.95. 2022 – Expected **Sep. 2026**

University of California San Diego, M.S. in **Mechanical Eng.,** GPA 3.76 2018 –2020

National Cheng Kung University, B.S. in **Civil** with minor of **Mechanical Eng.,** GPA 3.42 2013 – 2017

Technical Skills

- **Robotics Tools:** ROS, Embedded systems, UR, ABB
- **Programming & Dev.:** Python, C++, MATLAB, PyTorch, Shell, Git, Linux
- **Algorithms & Methods**: MPC, LQR, PID, Machine Learning/RL, Data-driven Inverse Dynamics, Motion Planning, Manipulation

Professional Experience

Autonomous Vehicle Engineer, Industrial Tech. Research Institution (ITRI), Taiwan Oct. 2020 – June 2022

- Optimized vehicle braking decision logic in ROS/C++, achieving a 20% reduction in unnecessary AEB engagements while meeting safety and comfort specifications, validated across 100+ mixed-traffic public road trials
- Troubleshot ADAS drive-by-wire controllers and enhanced CAN and digital I/O signal integrity via filtering and signal processing

Robotics Intern, Pacific Northwest National Laboratory (PNNL), WA June 2024 – Sep. 2024

- Delivered a 40% speed improvement in vision-based battery inspection systems using deep learning–driven process optimization
- Architected a multi-robot ROS system integrating 2 UR5 arms and 2 autonomous mobile robots with 10+ custom nodes, enabling end-to-end autonomous lab workflows

Intelligent Automation Intern, Boeing, WA June 2025 – Sep. 2025

- Achieved strong precision-recall performance (PR-AUC 0.85) for real-time drill anomaly detection using variational autoencoder on multi-dimensional industrial sensor data
- Delivered automated monitoring solutions that reduced inspection delays and improved quality control in aircraft wing assembly

Research Experience

Research Assistant, Precision Controls Lab, UW *(P2, P3)* Sep.2022 – Present

- Improved robotic platoon longitudinal tracking performance by 85% using cohesive network dynamics, achieving 45% faster task completion in 30+ human teleoperation trials with visual-servoing on TurtleBot3 (ROS/C++/Python)
- Enabled 95% obstacle avoidance rate for a unicycle robot by integrating GPU-accelerated MPPI with Control Barrier Functions
- Boosted inverted cart-pendulum tracking by 20% using a neural network-based inverse model for feedforward control
- Optimized robot trajectory of a 2-agent Game Dynamics using Sequential Quadratic Programming (SQP) approach

Research Assistant, Boeing Advanced Research Collaboration (BARC), UW *(P4, P5)* Sep.2022 – Present

- Improved large-scale aircraft assembly precision by 500% by developing a data-driven shape control system using ridge regression in MATLAB/Python, validated through simulation and industrial-grade experiments
- Reduced material forming process time by 100% by designing an online learning algorithm for time-varying material modeling using multiple ABB robot arms and camera feedback

Research Assistant, Gravish Lab, UCSD *(P1)* Mar. 2019 – June 2020

- Improved legged robot obstacle avoidance rate by 70% on rough terrain by developing an impedance controller without re-planning
- Reduced robot leg unsprung mass by 15% using Python-based convex optimization, improving agile feedback

Publications & Patents

P1. **Henry C.** et al. "Anisotropic Compliance of Robot Legs Improves Recovery from Swing-Phase Collisions" *(2021) Bioinsp. Biomim.*

P2. **Henry C.** et al. "Human-Led Decentralized Constant-Spacing Robot Platoons without Communication" *(2026) IEEE Robotics and Automation Letters*, under review

P3. Gombo, Y. et al. (incl. **Henry C.**) "Delayed Self-Reinforcement to Reduce Deformation during Decentralized Flexible-Object Transport" *(2023) IEEE Transections on Robotics.*

P4. Safwat, M. et al. (incl. **Henry C.**) "Data-Enabled Stochastic Iterative Shape Control for Assembly of Flexible Structures." *(2026) ASME Letters in Dynamics Systems and Control*

P5. Tatar, M. et al. (incl. **Henry C.**) "Manufacturing Systems and Methods for Shaping and Assembling Flexible Structures" (2025) U.S. Patent App. 2025/0322119.

Teaching Experience

Teaching Assistant, Junior level course: **ME373 & ME374 System Dynamics (1) (2)**, UW Winter 2021 – Spring 2022

- Conducted weekly 6-hour recitation sessions & office hour for 180 students, offering short lectures to address their questions